Olive oil polyphenols decrease blood pressure and improve endothelial function in young women with mild hypertension.

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BACKGROUND: Olive oil polyphenols have been associated with several cardiovascular health benefits. This study aims to examine the influence of a polyphenol-rich olive oil on blood pressure (BP) and endothelial function in 24 young women with high-normal BP or stage 1 essential hypertension. METHODS: We conducted a double-blind, randomized, crossover dietary-intervention study. After a run-in period of 4 months (baseline values), two diets were used, one with polyphenol-rich olive oil (~30 mg/day), the other with polyphenol-free olive oil. Each dietary period lasted 2 months with a 4-week washout between diets. Systolic and diastolic BP, serum or plasma biomarkers of endothelial function, oxidative stress, and inflammation, and ischemia-induced hyperemia in the forearm were measured. RESULTS: When compared to baseline values, only the polyphenol-rich olive oil diet led to a significant ($P < 0.01$) decrease of 7.91 mm Hg in systolic and 6.65 mm Hg of diastolic BP. A similar finding was found for serum asymmetric dimethylarginine (ADMA) (-0.09 ± 0.01 µmol/l, $P < 0.01$), oxidized low-density lipoprotein (ox-LDL) (-28.2 ± 28.5 µg/l, $P < 0.01$), and plasma C-reactive protein (CRP) (-1.9 ± 1.3 mg/l, $P < 0.001$). The polyphenol-rich olive oil diet also elicited an increase in plasma nitrites/nitrates ($+4.7 \pm 6.6$ µmol/l, $P < 0.001$) and hyperemic area after ischemia ($+345 \pm 386$ perfusion units (PU)/sec, $P < 0.001$). CONCLUSIONS: We concluded that the consumption of a diet containing polyphenol-rich olive oil can decrease BP and improve endothelial function in young women with high-normal BP or stage 1 essential hypertension.